

7	(a) (i)	oscillations are <u>damped</u> /amplitude decreases	B1	
		as magnet moves, flux is cut by coil	B1	
		e.m.f./current is induced in the coil	B1	
		causing energy loss in load OR force on magnet	B1	
		energy is derived from oscillations of magnet		
		OR force opposes motion of magnet	B1	[5]
	(ii)	$T = 0.60 \text{ s}$	C1	
		$\omega_0 (= 2\pi/T) = 10.5 \text{ rad s}^{-1}$	A1	[2]
	(b)	sketch: sinusoidal wave with period unchanged or slightly smaller	M1	
		same initial displacement, less damping	A1	[2]
	(c) (i)	sketch: general shape – peaked curve	M1	
		peak at ω_0 and amplitude never zero	A1	[2]
	(ii)	<u>resonance</u>	B1	[1]
	(iii)	useful: e.g. child on swing, microwave oven heating	B1	
		avoid: e.g. vibrating panels, vibrating bridges (for credit, stated example must be put in context)	B1	[2]

Section B

8 (a) e.g. infinite (voltage) gain